Group and solo flight: comparisons

What differs between group time and solo flight, and what remains when the controls are in place?

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Abstract

Group time differs from solo flight, but so do the pilots, the places and the routes of groups. This report compares paragliders in three populations, solo flights, group time and the same flights alone, and gives for each result the number of flights or pieces behind it. Raw, group time progresses straighter over a quarter of an hour than solo flights, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ of 0.893 against 0.837, and glides faster and flatter. On the long pieces group time keeps this progress to lags of hours, and it is straighter with more pilots in the group. The ordering holds in 5 of the 6 zones of 50 km with the most group flights. Against the same route flown on another day by isolated solo flights of the same wing level, the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from $+0.058$ to $+0.017$ [$+0.011$, $+0.025$] on 15 539 pieces: the route explains 70% of it, for experienced pilots and beginners alike. What remains is small and goes with flying together; whether the group causes it stays open.

1 Aim

Does a paraglider move differently in a group than alone? This report answers with a series of comparisons of group time with solo flight. Each has its own control, and a result is read together with the number of flights or pieces behind it, given in every table and figure.

All the populations are paragliders. Section 2 defines them, counts them and maps them, Section 3 compares them as they are, over lags up to hours and by the size of the group, Section 4 repeats the comparison in the zones with the most group flights, and Section 5 compares group time with the same route flown alone.

2 Populations and measures

Which flights, which stretches of flight, and how many of them?

Groups. The labels are those of the note *Group flights in free flight* (`methods/group-identification`). Two pilots are in contact when one is on the path of the other within $\delta = 2$ min, $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically (headings compared only in glide) for at least $\tau^* = 38.1$ min. The configurations of an instant are the connected components of the contact graph, and a group keeps the same members over a maximal interval. A contact means shared air within sight; it does not prove interaction.

States. After its *departure*, the first fix 3 km from the takeoff, every instant of a flight is in a group or solo; before it the pilot counts as neither. Flights that never depart are left out. We use paraglider flights logged every 10 s or finer, so that the labels exist at every instant; the phases (climb, search, glide) exist on the flights logged every second.

Populations. Three, all of paragliders.

- S, *solo flights*: flights that never entered a group, the whole of their time after the departure. They may have had contacts shorter than τ^*.
- G, *group time*: the stretches in a group of the flights that entered one at least once. A stretch stays in one group; G counts a flight once however many groups it joins.
- A, *the same flights alone*: the solo stretches of those same flights, a control with the same pilots, gliders, loggers and days as G.

The control of Section 5 is stricter than S and is defined there.

Measure. A *piece* is a stretch within one segment of the track, of at least 1250 s, on a 10-s grid of its clock. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta^2_f}(\tau) = \frac{1}{N_f(\tau)} \sum_t \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2, \tag{1}$$

with t and $t+\tau$ in the same piece. The ensemble MSD(τ) is its mean over the flights, one weight per flight, and H is half the slope of log MSD against log τ over the lags from 10 to 1000 s: $H = 1$ for a straight line, $H = \frac{1}{2}$ for a random walk. $H_{\text{loc}}(\tau)$ is the same slope over the lags within ± 0.25 decades of τ. It is read at $\tau = 1000$ s, over one to two climb–glide cycles. Climbs and glides are summarised by the share of time in climb, the share of searches that end in a climb, the climb rate, the glide speed and the glide ratio, each a ratio of sums over the phase runs of a population. Bands are 5–95% of bootstrap draws that resample the pilots and the dependency blocks of the grouping.

	solo flights S	group time G	same flights alone A
flights, H and H_{loc}	119 129	33 435	29 611
pieces of at least 1250 s	125 997	45 476	47 577
flights, time in climb and searches	80 278	24 099	23 469
flights, climb rate and glides	115 629	33 406	30 703

Table 1: Members of the populations, paragliders. The flights of G and A are the same flights, and A keeps those with a solo stretch of at least 1250 s.

Sizes. Table 1 counts the members. G has 33 435 flights in 45 476 pieces and S has 119 129 flights in 125 997. A has 29 611 flights in 47 577 pieces, fewer flights than G: the others have no solo stretch long enough. The phase statistics rest on fewer flights, those logged every second.

Where. Figure 1 maps the middle of the pieces of S and of G.

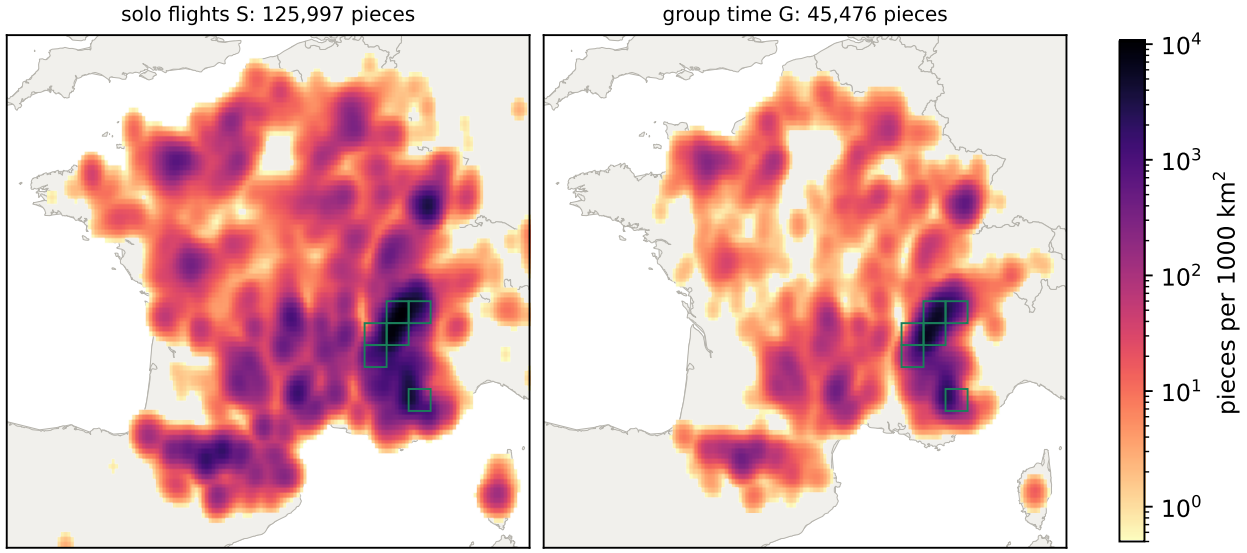


Figure 1: Density of the middles of the pieces of solo flights S (125 997 pieces, 119 129 flights) and of group time G (45 476 pieces, 33 435 flights), in pieces per 1000 km^2, on one colour scale. Squares: the zones of Section 4.

3 The raw difference

How do the three populations differ before any control?

Group time progresses straighter over long lags than solo flights: $H_{\text{loc}}(1000\text{ s})$ is 0.893 in G, 0.860 in A and 0.837 in S (Figure 2). It also spends more time climbing, ends more searches in a climb and glides faster and flatter, with a glide ratio of 12.9 against 10.4. The same flights alone sit between the two at long lags and

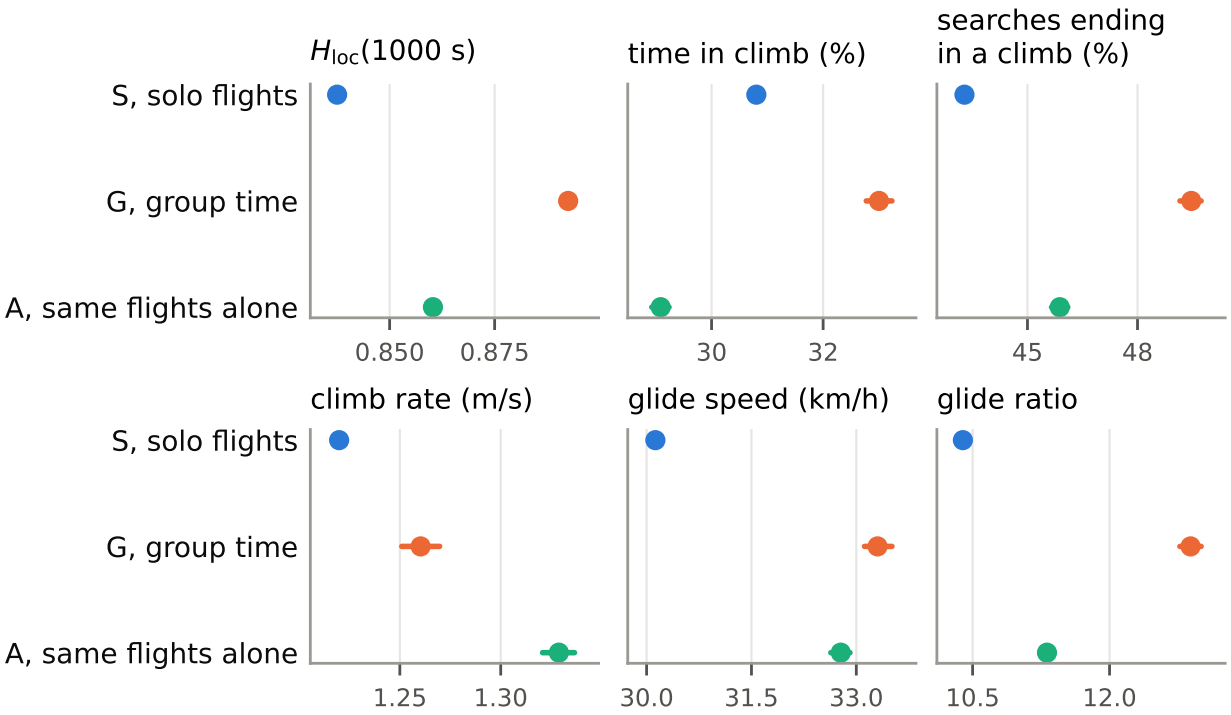


Figure 2: Pooled statistics of S, G and A with 5–95% bands. Exponents on 119 129, 33 435 and 29 611 flights, phase statistics on the flights of Table 1. Values in Table 2.

have the highest climb rate. Part of the raw difference may belong to who flies in groups, and to where and when they fly.

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	119 129	33 435	29 611
H (10–1000 s)	0.899 [0.898, 0.899]	0.916 [0.915, 0.917]	0.914 [0.913, 0.915]
H_{loc}(1000 s)	0.837 [0.837, 0.838]	0.893 [0.891, 0.894]	0.860 [0.859, 0.862]
time in climb (%)	30.8 [30.7, 30.9]	33.0 [32.8, 33.2]	29.1 [28.9, 29.2]
searches ending in a climb (%)	43.3 [43.2, 43.4]	49.5 [49.2, 49.7]	45.9 [45.7, 46.1]
climb rate (m/s)	1.22 [1.22, 1.22]	1.26 [1.25, 1.27]	1.33 [1.32, 1.34]
glide speed (km/h)	30.1 [30.1, 30.2]	33.3 [33.1, 33.5]	32.8 [32.6, 32.9]
glide ratio	10.4 [10.4, 10.4]	12.9 [12.8, 13.0]	11.3 [11.2, 11.4]
cross-country speed (km/h)	17.2 [17.2, 17.3]	20.0 [19.8, 20.3]	20.0 [19.8, 20.1]

Table 2: Pooled statistics of the populations of Table 1, with 5–95% bands. The first row is the number of flights of the exponents; the other rows rest on the flights with phases (Table 1).

3.1 Longer lags and the size of the group

How do the curves go over lags of hours, and with the number of pilots of the group?

A piece of at least 1250 s gives the lags up to 1000 s. A longer lag τ needs pieces of at least $\tau/0.8$ s, so a curve to a long lag is measured on the long pieces of its population only, and the population does not change along it. The populations of one panel of Figure 3 share the longest lag T, the longest at which every one of them keeps at least 100 flights, and are measured on the pieces of at least $T/0.8$ s (dashed); the thin lines are the curves of all the pieces up to 1000 s. The last panel gives the flights left at each lag, and Table 3 the flights of each curve. H_{loc} is read from MSD over the lags within ± 0.25 decades.

Pooled. On the long pieces group time stays straight longer. At 5 080 s H_{loc} is 0.890 [0.825, 0.931] in G, against 0.789 in S and 0.811 in A; at 9 350 s it is 0.805 [0.655, 0.881] against 0.597 and 0.622. For every population H_{loc} falls with the lag as the motion saturates, and A follows S more than G. The long series of G rests on few flights (Table 3), and its band is wide.

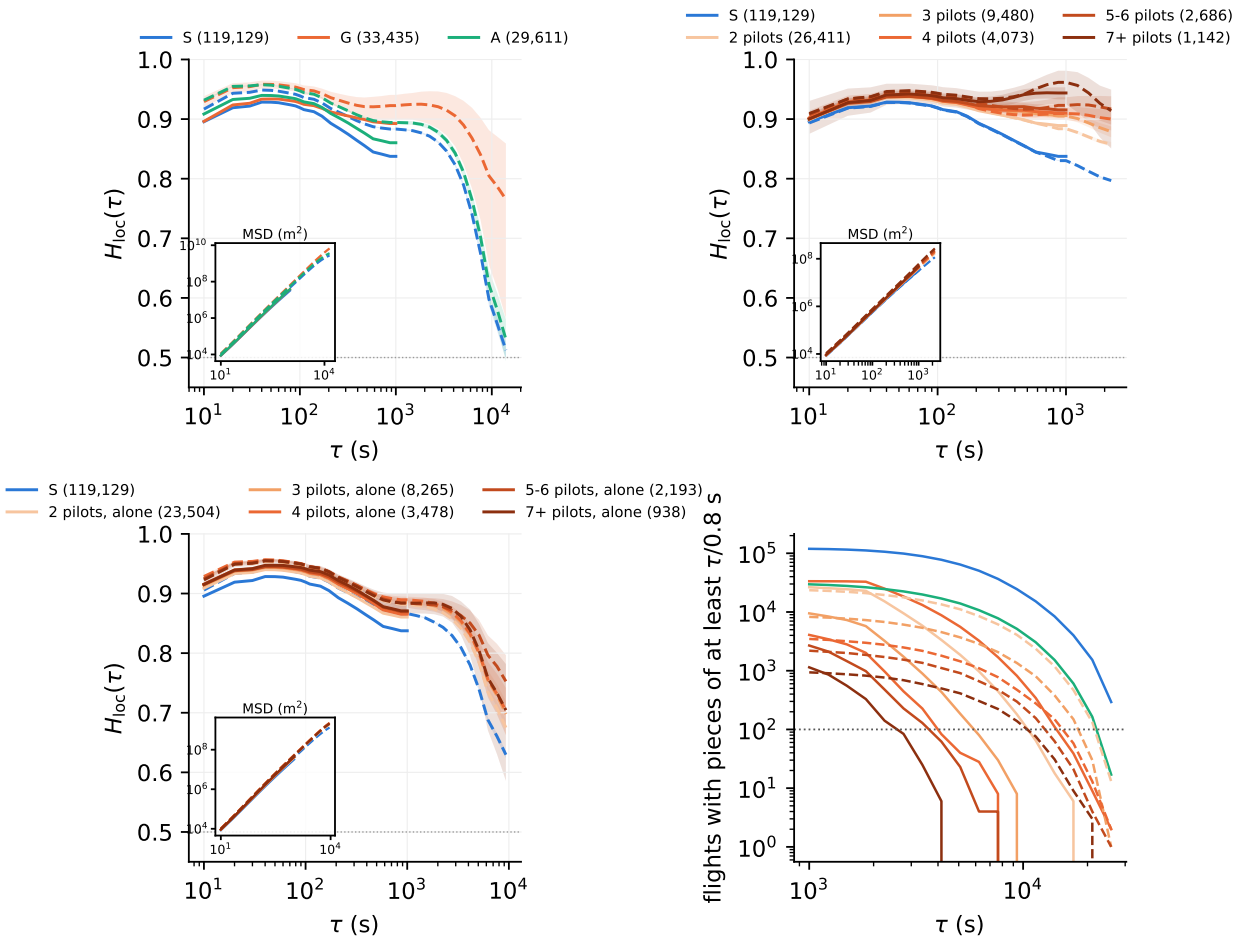


Figure 3: $H_{loc}(\tau)$ with an inset of MSD(τ) in m^2. Top left: S, G and A, lags up to 14 030 s. Top right: S and group time by the number of pilots of the group, up to 2 250 s. Bottom left: S and the same flights alone, by the size of the group the flight had, up to 9 350 s. Thin: all the pieces, up to 1000 s, with the flights in the legend; dashed: the pieces long enough for the longest lag of the panel; 5–95% bands. Bottom right: flights with pieces of at least $\tau/0.8$ s; dotted, the minimum of 100 flights.

population	flights up to 1000 s	flights of the long series
pooled populations, lags up to 14 030 s		
solo flights S	119 129	8 427
group time G	33 435	111
same flights alone A	29 611	1 526
group time by the size of the group, lags up to 2 250 s		
solo flights S	119 129	106 241
G, 2 pilots	26 411	15 181
G, 3 pilots	9 480	3 222
G, 4 pilots	4 073	973
G, 5–6 pilots	2 686	478
G, 7+ pilots	1 142	141
the same flights alone, by size, lags up to 9 350 s		
solo flights S	119 129	24 497
A, flights of 2 pilots	23 504	4 080
A, flights of 3 pilots	8 265	1 354
A, flights of 4 pilots	3 478	474
A, flights of 5–6 pilots	2 193	298
A, flights of 7+ pilots	938	139

Table 3: Flights of each curve of Figure 3: those of all the pieces, up to 1000 s, and those of the long series, with pieces of at least $T/0.8$ s for the longest lag T of the panel.

By size. Group time is straighter with more pilots, up to 5–6: at 2 250 s H_{loc} is 0.797 in S and 0.858, 0.879, 0.900, 0.917 and 0.914 [0.851, 0.949] for 2, 3, 4, 5–6 and 7+ pilots. The 7+ pilots, with few flights, are as straight as 5–6 within the band.

Alone, by size. The solo time of the flights of groups is above solo flights at every size: at 9 350 s 0.675 for flights that had groups of two, 0.753 for 5–6 pilots, against 0.630 in S; the band of the 7+ pilots, [0.586, 0.782], reaches S.

What we keep. A difference to explain, largest at long lags and in the glides: group time against solo flights, with the same flights alone in between; on the long pieces group time keeps its progress to lags of hours, and group time is straighter with more pilots up to 5–6. The next sections ask where the difference comes from.

4 The zones with the most group flights

Do the three populations differ in the same way where groups are most frequent?

A zone is a cell of 50 km × 50 km that holds the middle of a piece, whichever the takeoff of the flight; the pilots of a zone need not take off from the same place. The 6 cells with the most flights that have group time in them are kept, and each is named by the takeoff most flown by its group flights (the cells are on a fixed grid, so a busy site can fill two of them; Figure 1). In each zone the pieces of S, G and A are those of Section 2, the measure is the same, and a population is drawn when it has at least 20 flights in the zone. Bands are 5–95% of bootstrap draws that resample the dependency blocks of the grouping.

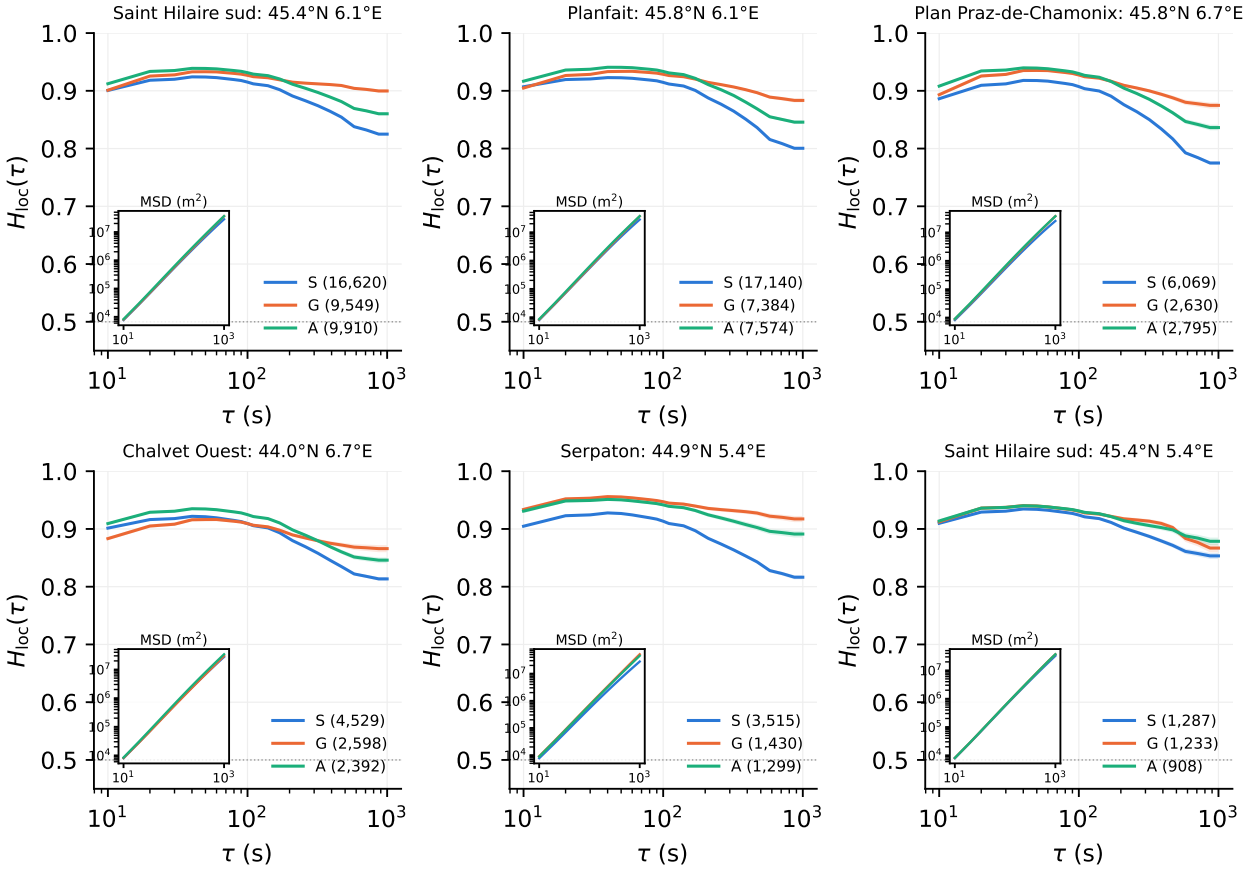


Figure 4: $H_{\mathrm{loc}}(\tau)$ of S, G and A in the 6 zones, with the 5–95% band and, inset, MSD(τ) in m² (log–log). The number of flights of each population is in the legend and in Table 4.

In 5 of the 6 zones $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is highest in group time, then in the same flights alone, then in solo flights, the order of the pooled populations (Figure 4, Table 4). In the other zone the same flights alone are above group time at 1000 s.

What we keep. The ordering of the three populations at long lags is there in most of the zones where groups fly most, and not only in the pooled data.

zone	group flights	flights S	flights G	flights A	$H_{\mathrm{loc}}(1000\ \mathrm{s})$ S	G	A
Saint Hilaire sud (45.4°N, 6.1°E)	9 549	16 620	9 549	9 910	0.825	0.900	0.860
Planfait (45.8°N, 6.1°E)	7 384	17 140	7 384	7 574	0.800	0.883	0.846
Plan Praz-de-Chamonix (45.8°N, 6.7°E)	2 630	6 069	2 630	2 795	0.775	0.875	0.836
Chalvet Ouest (44.0°N, 6.7°E)	2 598	4 529	2 598	2 392	0.814	0.866	0.846
Serpaton (44.9°N, 5.4°E)	1 430	3 515	1 430	1 299	0.816	0.917	0.891
Saint Hilaire sud (45.4°N, 5.4°E)	1 233	1 287	1 233	908	0.853	0.867	0.879

Table 4: The zones, by the number of flights with group time in them. Flights of each population and $H_{\mathrm{loc}}(1000\ \mathrm{s})$; the bands are in Figure 4.

5 The same route flown alone

Groups may form on a ridge or a convergence line where any pilot progresses straight, and the straightness would then belong to the route and not to the group. Does the gap remain when group time and its control fly the same route?

The control is the same route flown alone, on another day. It differs from the route test of the general report (`vs-solo/general`) on three points: the control may come from any site and date other than that of the piece, and every piece of group time that has a control is used, not a sample; the control must also have had nobody within 10 km at the same time, since a flight with no contact in the grouping can still have flown near others; and a piece is compared with controls of the same wing level, with the results given for experienced and beginner pilots apart.

5.1 The control and the measure

Which flights are the same route flown alone, and how is the gap with group time measured?

Matching. The pieces of group time are those of Section 2. A segment k of a solo flight is a control of the piece g when all hold.

(i) **Route.** At least 80% of the points of g, one every 30 s, lie within 500 m of the track of the flight of k, which flew them in the same direction; k is the part of the track between its points nearest to the first and to the last of them, and lasts at least 1250 s.

(ii) **Date.** Other than that of g, at any site.

(iii) **Season and hour.** The middles of g and k are in months within one (circular) and in local hours within two.

(iv) **Wing level.** The same: beginners (A, B) or experienced (C, D, CCC) in the FFVL catalogue.

(v) **Solo.** The flight never had a contact of any length, which is stricter than S: a contact shorter than τ^* already excludes it.

(vi) **Isolation.** At every point of k, one every 30 s, no other flight of the archive, alone or in a group, paraglider or hang glider, was within 10 km within one minute (the index of the archive holds one position per minute).

All the matches are kept. 19 786 pieces have a match under the first five conditions and 15 539 after the isolation; they have 85 524 matches, a median of 3 per piece, from 5 648 solo flights of 2 000 pilots, against 3 189 pilots on the group side.

Comparison. In analogy with the zones of the general report, the stratum is the piece. The control of a piece is the mean of the MSD of its matches, and each piece weighs the same on both sides. With n pieces and m_g the matches of the piece g,

$$\mathrm{MSD}_{\mathrm{G}}(\tau) = \frac{1}{n}\sum_{g}\mathrm{MSD}_{g}(\tau), \qquad \mathrm{MSD}_{\mathrm{ctrl}}(\tau) = \frac{1}{n}\sum_{g}\frac{1}{m_g}\sum_{k\in g}\mathrm{MSD}_{k}(\tau), \tag{2}$$

and both sides are means of many MSDs before their logarithm. The gap is $\Delta H_{\mathrm{loc}}(\tau) = H^{\mathrm{G}}_{\mathrm{loc}}(\tau) - H^{\mathrm{ctrl}}_{\mathrm{loc}}(\tau)$, positive when group time progresses straighter. The baseline compares the same pieces with the solo flights of their site-day and hour, so that the two gaps refer to the same group time; the share of the baseline gap that the route explains is

$$1 - \frac{\Delta H^{\mathrm{route}}_{\mathrm{loc}}(1000\ \mathrm{s})}{\Delta H^{\mathrm{base}}_{\mathrm{loc}}(1000\ \mathrm{s})}. \tag{3}$$

Uncertainty. Bands are 5–95% of 1000 bootstrap draws that resample the pilots and the dependency blocks of the grouping, independently, and recompute every statistic; each difference has its own band. The thin lines of the figures are the range when one year at a time is left out. The baseline and the explained share are point estimates.

What we keep. The control is the same route, on another day, in the same season, hour and wing level, flown by a solo pilot with nobody within 10 km. One gap, ΔH_{loc}, is read at 1000 s and compared with the baseline of the same pieces.

5.2 What remains of the gap

How much of the straighter progress of groups is left once the route is the same?

Over a quarter of an hour, group time still progresses straighter than the same route flown alone: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.017$ [$+0.011$, $+0.025$], against $+0.058$ for the same pieces against the solo flights of their site-day and hour (Table 5). Fixing the route lowers the gap by 70%, Eq. (3); its band stays away from zero. The long series, on the pieces that last long enough, gives about the same gap up to 3000 s, with bands wide enough to reach zero (Figure 5).

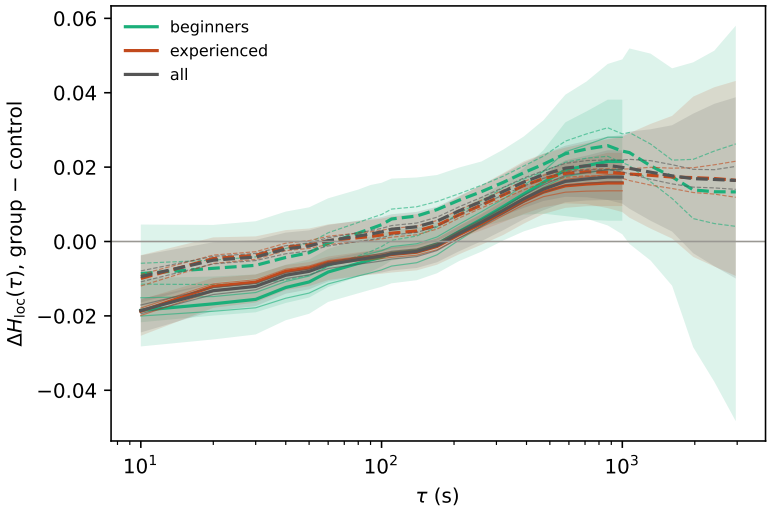


Figure 5: $\Delta H_{\mathrm{loc}}(\tau)$, group time minus the same route flown alone, for beginners (4 715 pieces), experienced pilots (10 824) and both (15 539). Continuous: the lags up to 1000 s; dashed: the long series, on the pieces that last long enough. Bands 5–95%, thin lines the range when one year is left out.

The gap changes sign with the lag: at short lags, within one climb or glide, group time is less straight than the same route flown alone, and it is more straight from about 200 s on (Figure 5).

wing level	pieces	matches	against site-day and hour	against isolated, same route	explained
all	15 539	85 524	$+0.058$	$+0.017$ [$+0.011$, $+0.025$]	70%
experienced	10 824	64 242	$+0.051$	$+0.016$ [$+0.010$, $+0.024$]	69%
beginners	4 715	21 282	$+0.068$	$+0.022$ [$+0.006$, $+0.038$]	68%

Table 5: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ of group time against the solo flights of the same site-day and hour, and against the isolated solo flights on the same route, with the 5–95% band; the share of the first that the route explains, Eq. (3).

What we keep. The route, with the season, the hour and the wing level, explains about 70% of the long-lag gap between group and solo flight. The rest, $+0.017$ at 1000 s, is small and goes with flying together.

5.3 Experienced pilots and beginners

Does the route explain the same share for both wing levels?

It does (Table 5, Figure 6). For the 10 824 pieces of experienced pilots $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from $+0.051$ to $+0.016$ [$+0.010$, $+0.024$], 69% explained. For the 4 715 pieces of beginners it falls from $+0.068$ to $+0.022$ [$+0.006$, $+0.038$], 68% explained. The beginners' baseline gap is larger and their band wider, since they have fewer pieces; the two levels differ by one point in the share explained, well within their bands.

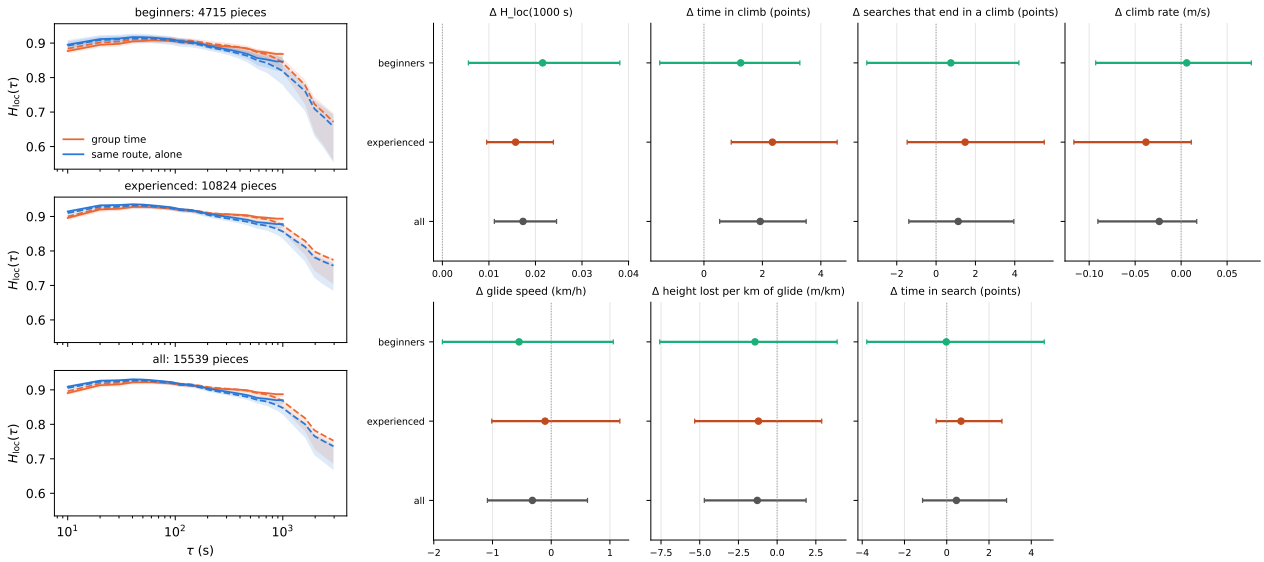


Figure 6: Left: $H_{\mathrm{loc}}(\tau)$ of group time (orange) and of the same route flown alone (blue), by wing level, with the number of pieces in the panel titles; dashed, the long series. Right: the difference group minus control of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ and of the statistics of climbs and glides, with 5–95% bands, by wing level.

What we keep. Experienced pilots and beginners give the same share explained, so that the route accounts for the same part of the gap whatever the wing level.

5.4 Climbs and glides

Does a group climb and glide differently from the same route flown alone?

Table 6 compares the time in each phase and the statistics of climbs and glides of the pieces that have phases (loggers of 1 s) with the same statistics on their controls. Group time has more time in climb and less in glide, and its pilots start more climbs per hour; the climb rate, the glide speed, the glide ratio and the height lost per kilometre of glide differ by less than their bands. In a group the pilot spends more time climbing, and the climb itself is not faster.

all levels	group time	same route alone	difference
pieces, segments	10 736	42 685	
time in glide (%)	57.5	59.9	−2.4 [−5.0, −1.0]
time in search (%)	13.7	13.2	+0.5 [−1.1, +2.8]
time in climb (%)	28.8	26.9	+1.9 [+0.5, +3.5]
searches ending in a climb (%)	49.6	48.5	+1.1 [−1.4, +4.0]
climbs per hour	7.46	6.99	+0.48 [+0.20, +0.94]
climb rate (m/s)	1.12	1.14	−0.02 [−0.09, +0.02]
glide speed (km/h)	28.6	28.9	−0.3 [−1.1, +0.6]
glide ratio	16.2	15.9	+0.3 [−0.5, +1.2]
height lost per km of glide (m/km)	61.6	62.8	−1.3 [−4.7, +1.9]

Table 6: Climbs and glides of group time and of the same route flown alone, both wing levels together, on 10 736 pieces of group time with phases and 42 685 matched segments; the difference group minus control with its 5–95% band. Times are shares of the time flown, in %, and their differences are in percentage points.

The right of Figure 6 gives the same differences for each level: the bands of the beginners are wider and contain zero for every statistic of the phases.

What we keep. On the same route a group spends more time climbing and less gliding; the speed, the rate and the glide ratio do not differ.

6 Conclusions and limits

What can be concluded, and what cannot?

What we keep.

- Raw, group time (G, 33 435 flights) progresses straighter than solo flights (S, 119 129) over long lags, and the same flights alone (A, 29 611) sit in between.
- On the long pieces group time keeps its progress to lags of hours, and is straighter with more pilots up to 5–6. The ordering of S, G and A is there in 5 of the 6 zones with the most group flights.
- The route, with the season, the hour and the wing level, explains 70% of the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ between group time and solo flight; the same for experienced pilots and beginners.
- What remains, $+0.017$ [$+0.011$, $+0.025$], is small, with a band away from zero; at short lags group time is less straight than the control.
- On the same route a group spends more time climbing and less gliding; the climb rate and the glide speed do not differ.

Limits.

- The raw comparison fixes nothing: S and G differ in pilots, places, days and hours.
- The days differ: season and hour are matched, the weather of the day is not.
- The controls flew with nobody within 10 km. They may differ from the pilots who fly in groups, and this test has no selection contrast to measure it.
- Only the 15 539 pieces whose route was flown alone on another day enter; their routes are frequented ones.
- The bands measure the sampling of pilots and blocks. The radius and the cover of the route were varied only in the general report, on a sample, and the isolation distance not at all.

Fixing the route removes most of the straighter progress of groups; whether what remains is caused by the group stays open.